KABIR KANHA ARORA
1710110165
CSD 205

classmate
Date 9/10/18
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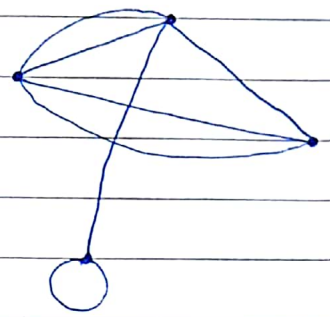
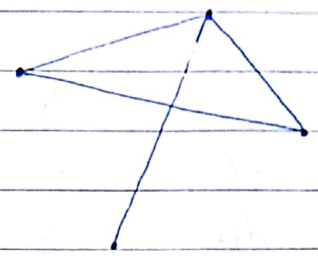
ASSIGNMENT
(Tutorial #6)

A1: A simple graph is one that does not have multiple edges (an edge that is identical to another, in the same graph) or loops (an edge from a vertex to itself)

eg:

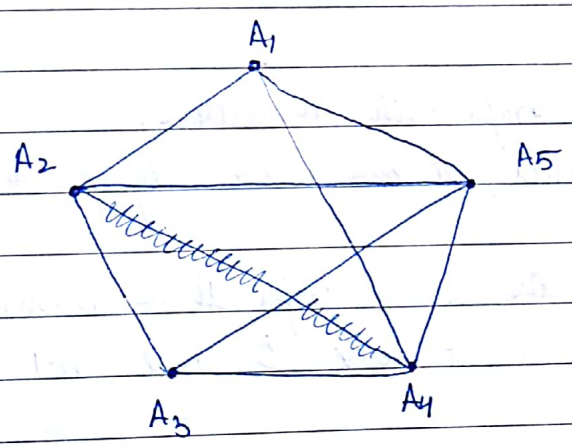


A non-simple graph

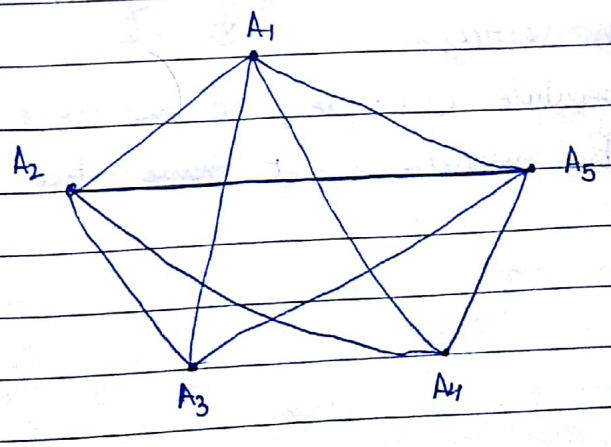


A simple graph

A2: a)



b)



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A3. No. of vertices = 8
No. of edges = 12

$d(a) \rightarrow 3$
$d(b) \rightarrow 2$
$d(c) \rightarrow 4$
$d(d) \rightarrow 0$
$d(e) \rightarrow 6$
$d(g) \rightarrow 4$
$d(h) \rightarrow 2$
$d(i) \rightarrow 3$

Isolated vertices → d ($d(d) = 0$)
~~Isolated~~ Pendant vertices → NONE (None of them have degree = 1)

A4: For a simple graph with n vertices,
for every vertex, it can have degree between 0 and n-1.
However, in the same graph, it is obvious that two vertices cannot have 0 and n-1 as their degrees.
Therefore, we have a range of n-1 degrees and n vertices.
Applying pigeonhole principle, it is clear that at least two vertices must have the same degree.

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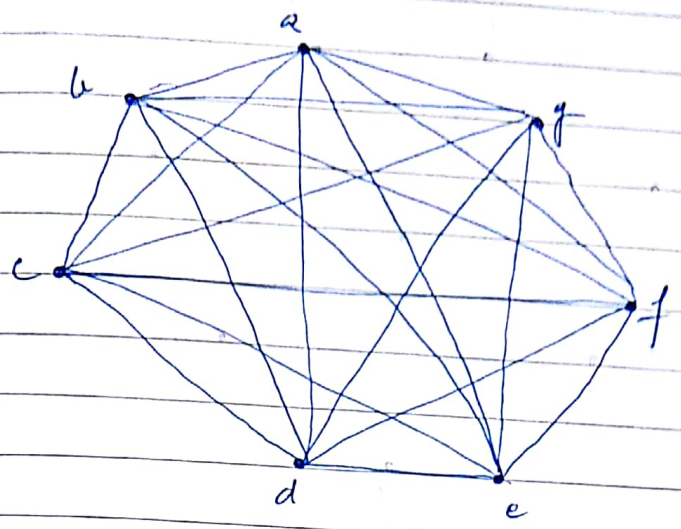
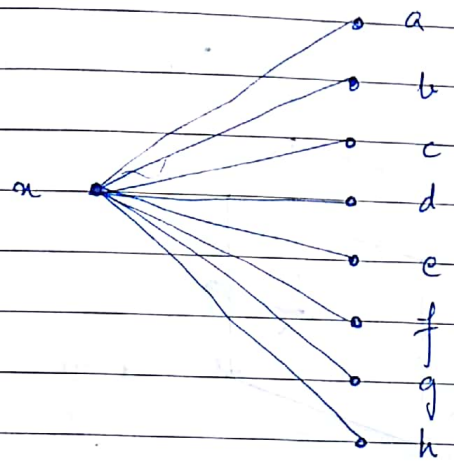
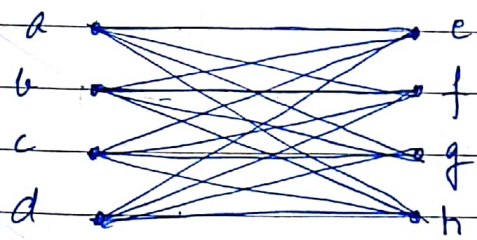
A5: (i) K_7



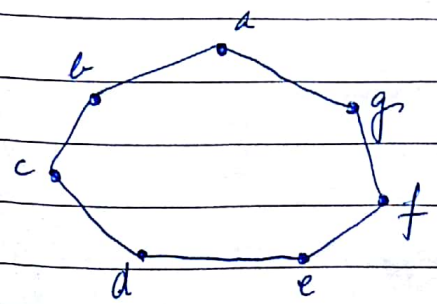
(ii) $K_{1,8}$



(iii) $K_{4,4}$



(iv) C_7



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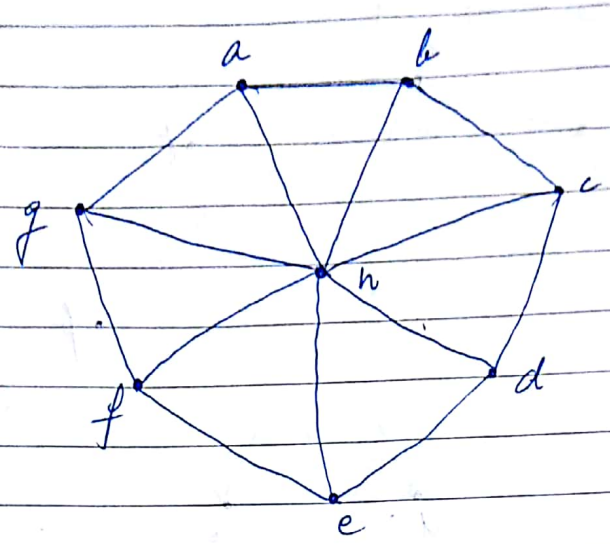
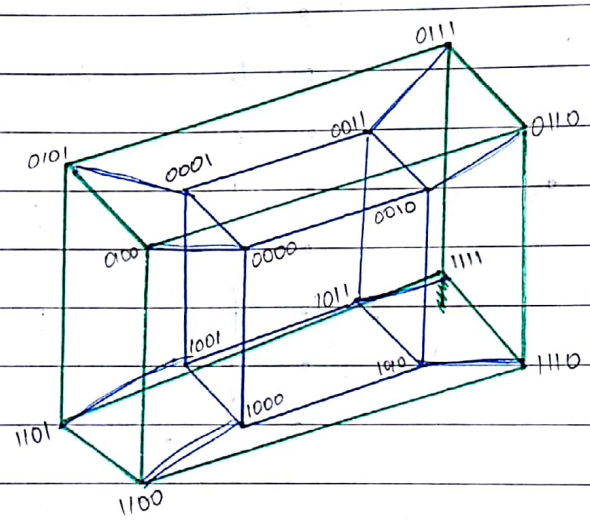
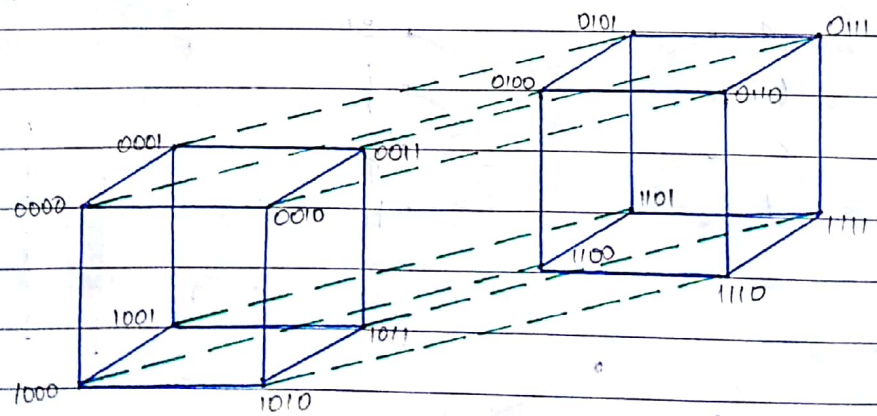
(v) W_7



(vi) Q_4



ALTER



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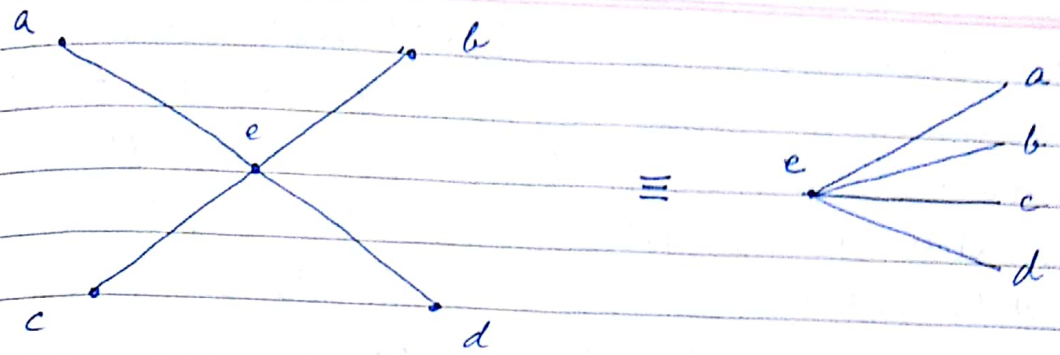
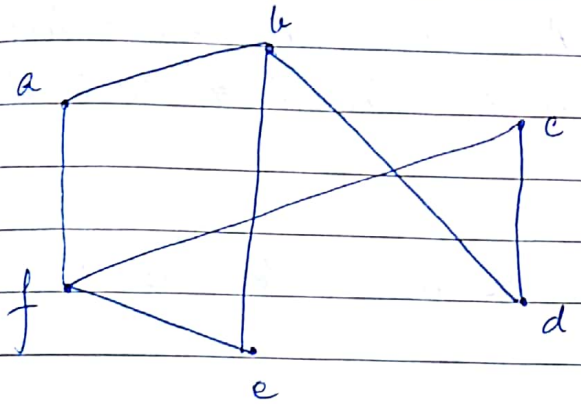
Ans: a)

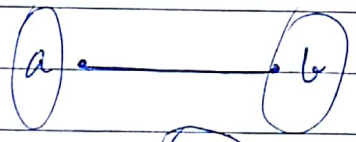


✓ BIPARTITE

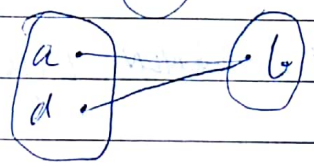
b)



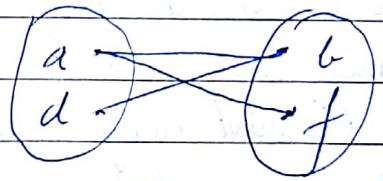
Edge b/w a-b So,



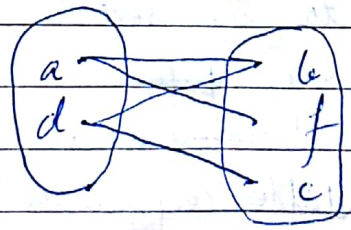
Edge b/w b-d So



Edge b/w a-f So,



Edge b/w c-d So,



BUT,

we see that there is also an edge b/w f and c in the graph, which means that the graph is <u>NOT BIPARTITE</u>.

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A7: Hall's theorem deals with a bipartite graph and gives us a necessary and sufficient condition for being able to select a matching that covers atleast one side of the graph.
The stable Marriage Problem is the problem of finding a stable matching b/w two sets of elements given an ordering of preferences for each element.

A matching is not stable if:
(i) There is an element A in one set which prefers B from the second set over the element to which it is already matched
AND,
(ii) B also prefers A over the element to which B is already matched.

ALGORITHM:

Assume that WOMEN propose to MEN

Step 1 → All women simultaneously propose to their 1st choices and the men reply with a "maybe" to the woman whom he prefers the most among those who've proposed to him and "NO" to the rest.

Step 2 → All women ~~propose~~ who are currently unengaged propose to their next choice and the process carries on until ~~at least~~ a stable configuration is found.
i.e:
(i) If all women get engaged, OR
(ii) If all women receive replies to all their proposals.

ASSUMPTIONS → ONLY HETEROSEXUAL MARRIAGES
→ EVERYONE WANTS TO GET MARRIED.

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A8: Let us say that the first K women on the island that we speak to are all willing to marry any man from a set of K men and hence those K men are willing to marry any of these K women.

Clearly, the $(K+1)^{th}$ woman cannot pick the same set of K men, because that would imply that the particular set of K men is willing to marry any of the (K+1) women, which VIOLATES the condition of exactly K.

Hence, there has to be at least one different man in the list of the $(K+1)^{th}$ woman and thus can be a stable pairing (match).

The same argument can be extended to all the remaining women and so, we do get a match for everyone on the island.

A9: (i) K_n

Vertices $\rightarrow n$

Edges $\rightarrow {}^nC_2$

(ii) C_n

Vertices $\rightarrow n$

Edges $\rightarrow n$

(iii) W_n

Vertices $\rightarrow n+1$

Edges $\rightarrow 2n$

(iv) $K_{m,n}$

Vertices $\rightarrow m+n$

Edges $\rightarrow m \times n$

(v) Q_n

Vertices $\rightarrow 2^n$

Edges $\rightarrow n \cdot 2^{n-1}$

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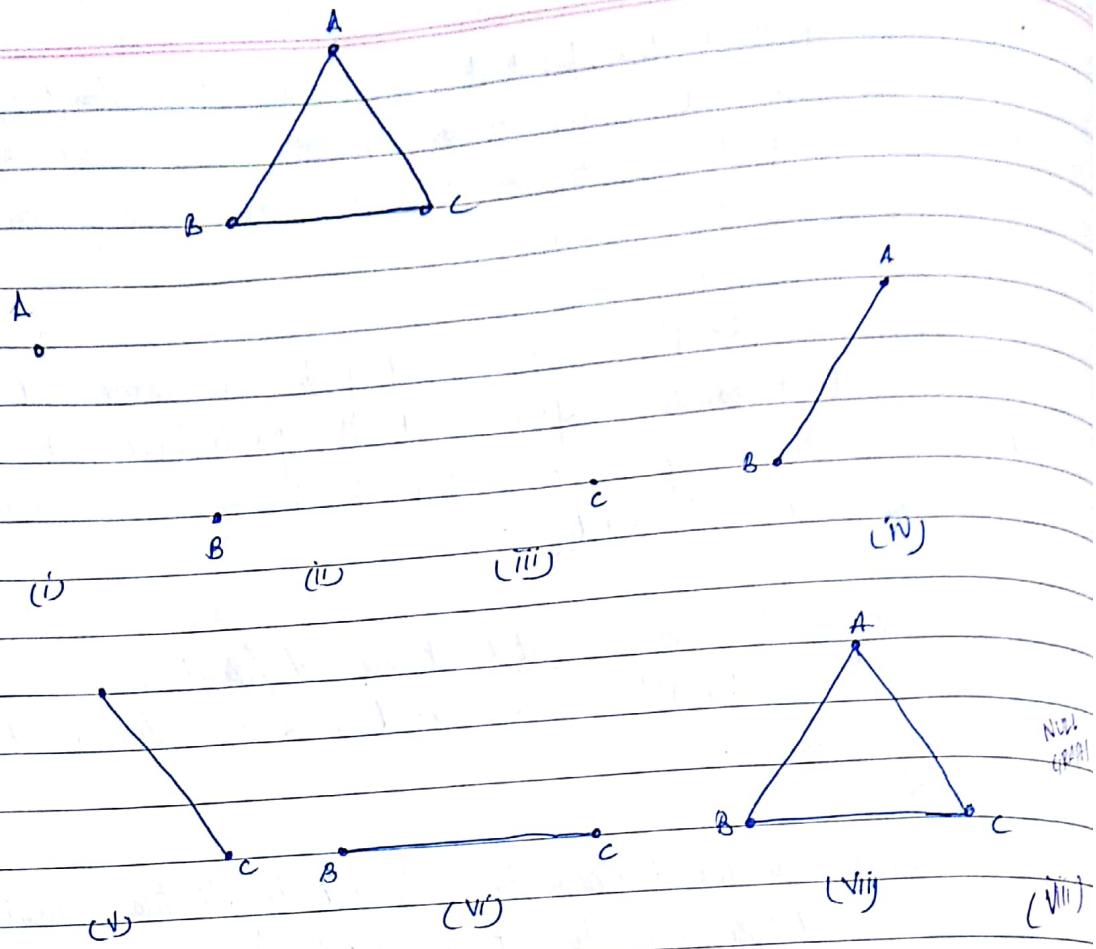
A10 :



Ans : 7

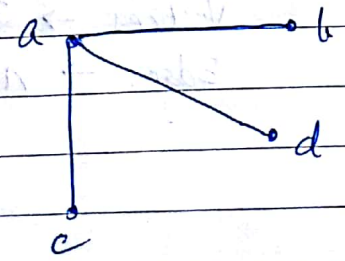
Verify :

$^3C_1 + {}^3C_2 + {}^3C_3 + {}^3C_0$

$= 3 + 3 + 1 + 1$

$= \underline{\underline{8}}$ ✓

A11:



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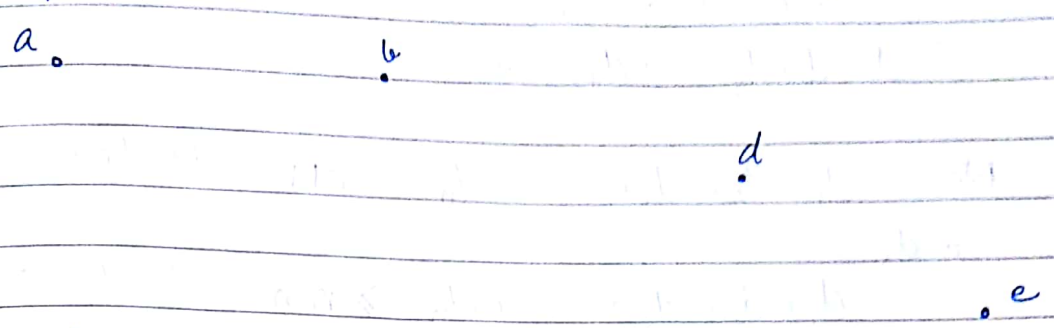
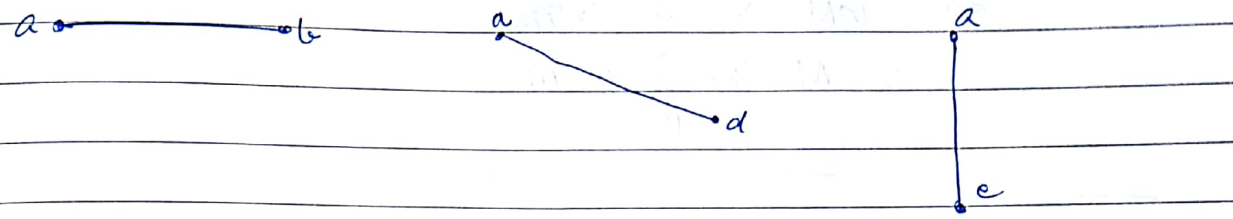
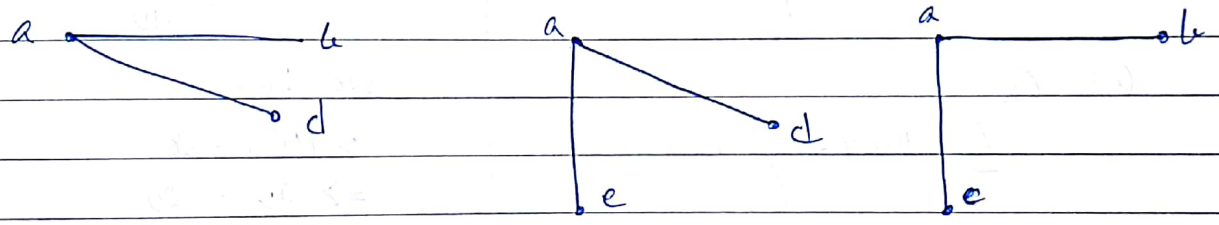
Subgraphs →



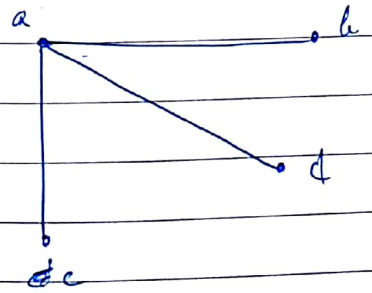
(i) (ii) (iii) (iv)



(v) (vi) (vii)



(viii) (ix) (x)



NULL GRAPH

(xi) (xii)

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A12: We know from Handshake Lemma, that

$$\Sigma d(v) = 2e$$

$$\therefore d_1 + d_2 + d_3 + \dots + d_n = 2e$$

Now $d_1 + d_2 + d_3 + \dots + d_n \leq nM$ → Max degree

and,

$$d_1 + d_2 + d_3 + \dots + d_n \geq nm$$ → Min degree.

$$\Rightarrow vM \geq \Sigma d(v) \geq vm$$

$$\Rightarrow vM \geq 2e \geq vm$$

$$\therefore M \geq \frac{2e}{v} \geq m$$

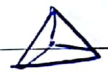
A13: (a) (i) K_n

All values of n

(ii) C_n

All values of n

(iii) W_n



~~All values of~~ $n = 3$

(iv) Q_n

All values of n

(c) $d(v) = 4$

$\Sigma d(v) = 4 \times v$

$= 4v$

We know

$\Sigma d(v) = 2e$

$\Rightarrow 4v = 20$

$\Rightarrow v = 5$

b) $K_{m,n}$

Degree of all vertices in $m = n$

Degree of all vertices in $n = m$

$\therefore$ For a regular graph

$m = n$

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